



BMW

Sales Analytics Dashboard

An end-to-end Data Analytics Project

Prepared By:

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Tools Used:

Python, PostgreSQL, Power BI, Git/GitHub

Introduction

About the project

The **BMW Sales Analytics Dashboard** is a Business Intelligence portfolio project created to analyze BMW sales data from **2010 to 2024**. The aim of this project is to transform raw sales data into meaningful insights using data analytics and interactive visualizations.

The project covers the complete analytics workflow, including understanding the dataset, cleaning and preparing the data, performing SQL analysis, and building an interactive dashboard in Power BI. The dashboard allows users to explore sales trends, compare BMW models, analyze regional performance, and study vehicle specifications through interactive visuals and KPIs.

The dashboard is designed with a clean and modern interface inspired by BMW's design language while being developed independently as a portfolio project for learning and demonstration purposes.

Project Overview

Item	Details
Project Name	BMW Sales Analytics Dashboard
Project Type	Business Intelligence Portfolio Project
Industry	Automotive
Dataset Source	Kaggle
Data Period	2010–2024
Dataset Size	50,000 records
Tools Used	Python, SQL, Power BI, Git & GitHub
Dashboard Pages	7

Business Understanding

Business Problem

Businesses generate large amounts of sales data every year, but raw data alone does not provide meaningful insights. To make informed business decisions, this data needs to be analyzed and presented in a clear and interactive way.

For this project, the goal is to design a Business Intelligence dashboard using historical BMW sales data from 2010 to 2024. The dashboard should help users understand sales trends, compare the performance of different BMW models, analyze regional sales, and explore vehicle characteristics such as fuel type, transmission, engine size, and pricing.

The challenge is to transform a large dataset into an easy-to-understand dashboard that supports data-driven decision-making through meaningful visualizations and interactive analysis.

Business Objectives

The main objectives of this project are:

- Analyze BMW sales performance between 2010 and 2024.
- Identify the best-performing BMW models based on sales volume.
- Compare sales performance across different regions.
- Study the relationship between vehicle specifications and pricing.
- Analyze trends based on fuel type, transmission, and engine size.
- Build an interactive Power BI dashboard with dynamic filtering and navigation.
- Present business insights in a clear and visually appealing manner.
- Demonstrate practical skills in Python, SQL, and Power BI through an end-to-end analytics project.

KPIs

KPI	Description
Total Sales Volume	Total number of vehicles sold
Average Vehicle Price	Average selling price of BMW vehicles
Total BMW Models	Number of unique models available in the dataset
Total Regions	Number of regions represented in the dataset
Average Engine Size	Average engine size (L)
Average Mileage	Average mileage (KM)
Top Selling Model	Model with the highest sales volume
Top Performing Region	Region with the highest sales volume
Most Preferred Fuel Type	Fuel type with the highest sales
Most Common Transmission	Transmission type most frequently recorded

Requirement

Requirement	Description
Interactive Filters	Filter data by Year, Model, Region, Fuel Type, and Transmission
KPI Cards	Display key business metrics
Cross Filtering	Selecting one visual updates related visuals
Interactive Charts	Allow users to explore trends and comparisons
Navigation	Easy movement between dashboard pages

Business Questions

- Which BMW model has the highest sales volume?
- Which region contributes the most to overall sales?
- How have sales changed from 2010 to 2024?
- Which fuel type is most popular?
- Which transmission type is most common?
- How does engine size influence vehicle price?
- Which models have the highest average selling price?
- What are the sales trends across different years?
- How are sales distributed across regions?
- Which vehicle characteristics are associated with higher sales?

Project Workflow

Workflow Overview

The BMW Sales Analytics Dashboard was developed by following a structured Business Intelligence workflow. The project began with understanding the business problem and collecting the dataset. The data was then prepared and validated using Python before being stored in PostgreSQL for SQL-based analysis. Finally, Microsoft Power BI was used to create an interactive dashboard, enabling meaningful business insights and recommendations.



Workflow Description

Phase	Description
Business Understanding	Defined the business problem, objectives, KPIs, and analytical requirements.
Data Collection	Collected the BMW Sales Dataset (2010–2024) from Kaggle.
Data Understanding	Explored the dataset structure, attributes, and overall data quality.
Data Preparation (Python)	Validated the dataset, checked missing values, duplicates, data types, and performed descriptive analysis using Pandas.
PostgreSQL Database	Created the database and imported the cleaned dataset for SQL analysis.
SQL Business Analysis	Executed business queries to analyze sales trends, regional performance, pricing, and model-wise insights.
Power BI Dashboard	Developed an interactive multi-page dashboard with KPIs, slicers, charts, and navigation.
Business Insights	Identified key trends and performance patterns from the dashboard.
Business Recommendations	Proposed recommendations based on the analytical findings.

Dataset Overview

Attribute	Details
Dataset Name	BMW Sales Data (2010–2024)
Source	Kaggle
Time Period	2010–2024
Number of Records	50,000
Number of Columns	11
File Format	CSV

Data Understanding

Missing values

The dataset was examined for missing values using the `isnull().sum()` function in Pandas. The analysis showed that **no missing values were present in any of the eleven columns.**

Observation

- ❖ Total missing values: **0**
- ❖ No imputation or missing value treatment was required.

```
Model          0
Year           0
Region         0
Color          0
Fuel_Type      0
Transmission   0
Engine_Size_L  0
Mileage_KM     0
Price_USD      0
Sales_Volume   0
Sales_Classification 0
dtype: int64
```

Duplicates

The dataset was checked for duplicate records using the `uplicated().sum()` function.

Observation

- ❖ Duplicate rows found: **0**
- ❖ The dataset contained only unique records.

```
np.int64(0)
```

Outliers

The numerical variables (Year, Engine_Size_L, Mileage_KM, Price_USD, and Sales_Volume) were reviewed using descriptive statistics and validation checks to identify any unusual or unrealistic values.

Observation

- ❖ No invalid numerical values were detected.

- ❖ All values fall within reasonable ranges based on the dataset.
- ❖ Therefore, no outlier treatment was performed.

	count	mean	std	min	25%	50%	75%	max
Year	50000.0	2017.01570	4.324459	2010.0	2013.00	2017.0	2021.00	2024.0
Engine_Size_L	50000.0	3.24718	1.009078	1.5	2.40	3.2	4.10	5.0
Mileage_KM	50000.0	100307.20314	57941.509344	3.0	50178.00	100388.5	150630.25	199996.0
Price_USD	50000.0	75034.60090	25998.248882	30000.0	52434.75	75011.5	97628.25	119998.0
Sales_Volume	50000.0	5067.51468	2856.767125	100.0	2588.00	5087.0	7537.25	9999.0

Initial observations

The initial exploration of the dataset revealed several important characteristics:

- ❖ The dataset contains **50,000 records** and **11 features**.
- ❖ The data covers the period **2010–2024**.
- ❖ Both categorical and numerical attributes are available for analysis.
- ❖ All columns contain complete information without missing values.
- ❖ The dataset is clean and well-structured for Business Intelligence analysis.

```
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Model                 50000 non-null  str
1   Year                  50000 non-null  int64
2   Region                50000 non-null  str
3   Color                 50000 non-null  str
4   Fuel_Type             50000 non-null  str
5   Transmission          50000 non-null  str
6   Engine_Size_L         50000 non-null  float64
7   Mileage_KM            50000 non-null  int64
8   Price_USD             50000 non-null  int64
9   Sales_Volume          50000 non-null  int64
10  Sales_Classification  50000 non-null  str
dtypes: float64(1), int64(4), str(6)
memory usage: 4.2 MB
```

Data quality assessment

Quality Check	Result
Missing Values	No missing values found
Duplicate Records	No duplicate records found
Data Types	All columns have appropriate data types
Numerical Validation	No invalid numerical values detected
Overall Data Quality	Excellent

Data Preparation

Overview

Before performing SQL analysis and dashboard development, the dataset was prepared to ensure it was suitable for analysis. Python (Pandas) was used to validate the dataset, confirm its quality, and verify that no preprocessing issues would affect the analytical results. Since the dataset was already well-structured, only validation and preparation steps were required before importing it into PostgreSQL.

Preparation Steps

Step	Description
Dataset Loading	Imported the CSV file into Python using Pandas.
Duplicate Validation	Verified that no duplicate records existed.
Missing Value Validation	Checked every column for missing values.
Data Type Validation	Confirmed that each column had the correct data type.
Numerical Validation	Reviewed numerical columns for unrealistic values.
Dataset Verification	Confirmed that the dataset was ready for SQL analysis.

Python Functions Used

Function	Purpose
pd.read_csv()	Load the dataset
df.info()	Check dataset structure and data types
df.describe()	Generate summary statistics
df.isnull().sum()	Identify missing values
df.duplicated().sum()	Detect duplicate rows
df.dtypes	Verify data types

Data Preparation Summary

Validation	Status	Action Taken
Missing Values	No Missing Values	No action required
Duplicate Records	No Duplicates	No action required
Data Types	Valid	No action required
Numerical Values	Valid	No action required
Dataset Ready	Yes	Imported into PostgreSQL

Conclusion

The data preparation process confirmed that the dataset was complete, consistent, and analysis-ready. No additional cleaning or transformation was required because the dataset contained no missing values, duplicate records, or invalid data types. The validated dataset

was then imported into PostgreSQL for SQL analysis and later connected to Microsoft Power BI for dashboard development.

Python Data Preparation

Overview

Python played an important role in the initial phase of this project by validating and preparing the dataset before it was imported into PostgreSQL for SQL analysis. The Pandas library was used to inspect the dataset, verify data quality, generate summary statistics, and ensure that the data was suitable for further analysis. Since the dataset was already well-structured, no major cleaning or transformation was required.

Objectives

The objectives of using Python in this project were:

- Load the BMW sales dataset into the Python environment.
- Examine the structure and contents of the dataset.
- Validate missing values and duplicate records.
- Verify the data types of all columns.
- Generate descriptive statistics for numerical attributes.
- Confirm that the dataset was ready for SQL analysis and dashboard development.

Libraries Used

Library	Purpose
Pandas	Data loading, validation, and manipulation
NumPy	Numerical operations and statistical support

Python Functions Used

Function	Purpose
pd.read_csv()	Import the dataset into Pandas
df.head()	Display the first few records
df.info()	Examine dataset structure and data types
df.describe()	Generate descriptive statistics
df.isnull().sum()	Check for missing values
df.duplicated().sum()	Detect duplicate records
df.dtypes	Verify data types of all columns

Python Code Summary

Load Dataset

```
# Load the Dataset
df = pd.read_csv("dataset/data_raw.csv")
```

[7] ✓ 1.0s

Dataset Information

```
df.info()
```

[5] ✓ 0.8s

```
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Model                  50000 non-null  str     
1   Year                   50000 non-null  int64   
2   Region                 50000 non-null  str     
3   Color                  50000 non-null  str     
4   Fuel_Type              50000 non-null  str     
5   Transmission           50000 non-null  str     
6   Engine_Size_L          50000 non-null  float64 
7   Mileage_KM             50000 non-null  int64   
8   Price_USD              50000 non-null  int64   
9   Sales_Volume           50000 non-null  int64   
10  Sales_Classification   50000 non-null  str     
dtypes: float64(1), int64(4), str(6)
memory usage: 4.2 MB
```

Missing Values

```
df.isnull().sum()
```

[8] ✓ 4.5s

Model	0
Year	0
Region	0
Color	0
Fuel_Type	0
Transmission	0
Engine_Size_L	0
Mileage_KM	0
Price_USD	0
Sales_Volume	0
Sales_Classification	0
dtype: int64	

Duplicate Records

```
df.duplicated().sum()
```

[14] ✓ 0.8s

np.int64(0)

Summary Statistics

```
df.describe()
```

[12] ✓ 0.7s

	Year	Engine_Size_L	Mileage_KM	Price_USD	Sales_Volume
count	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000
mean	2017.015700	3.247180	100307.203140	75034.600900	5067.514680
std	4.324459	1.009078	57941.509344	25998.248882	2856.767125
min	2010.000000	1.500000	3.000000	30000.000000	100.000000
25%	2013.000000	2.400000	50178.000000	52434.750000	2588.000000
50%	2017.000000	3.200000	100388.500000	75011.500000	5087.000000
75%	2021.000000	4.100000	150630.250000	97628.250000	7537.250000
max	2024.000000	5.000000	199996.000000	119998.000000	9999.000000

Output Summary

Metric	Result
Dataset Loaded	Successfully
Records Processed	50,000
Features	11
Missing Values	0
Duplicate Records	0
Data Types	Valid
Dataset Status	Ready for SQL Analysis

Key Outcomes

The Python validation process confirmed that the dataset was complete, consistent, and suitable for further analysis. No missing values, duplicate records, or incorrect data types were identified. The descriptive statistics provided an overview of the numerical variables, and the validated dataset was then imported into PostgreSQL for SQL-based business analysis.

SQL Analysis

Overview:

After validating the dataset using Python, the cleaned data was imported into PostgreSQL for business analysis. SQL was used to query the dataset, calculate key performance indicators (KPIs), and answer business questions related to sales performance, regional trends, pricing, and vehicle characteristics.

Objectives

- Import the prepared dataset into PostgreSQL.
- Perform business-oriented SQL analysis.
- Calculate KPIs using aggregate functions.
- Identify trends and patterns.
- Support Power BI dashboard development.

Database Design

Component	Description
Database	bmw_sales_data
Table	bmw_sales
Records	50,000
Columns	11

SQL Features Used

Feature	Purpose
SELECT	Retrieve data
WHERE	Filter records
GROUP BY	Aggregate data
ORDER BY	Sort results
COUNT()	Count records
SUM()	Total sales
AVG()	Average calculations
MAX()	Highest values
MIN()	Lowest values

SQL queries

Q1. How many sales records are available in the dataset

SQL Query

```
--Q1. Count the total number of sales records in the dataset  
SELECT COUNT(*) AS Total_Records FROM bmw_sales;
```

Output

	total_records bigint
1	50000

Q2. How many unique BMW models are available?

SQL Query

```
--Q2. Count the total number of unique BMW models available  
SELECT COUNT(DISTINCT Model) AS Total_Models FROM bmw_sales;
```

Output

	total_models bigint
1	11

Q3. Which BMW models have the highest number of sales records?

SQL Query

```
--Q3. Find the number of sales records for each BMW model  
SELECT Model, COUNT(*) AS Total_Sales FROM bmw_sales  
GROUP BY Model ORDER BY Total_Sales DESC;
```

Output

	model character varying (50)	total_sales bigint
1	7 Series	4666
2	i3	4618
3	i8	4606
4	3 Series	4595
5	5 Series	4592
6	X1	4570
7	X3	4497
8	X5	4487
9	X6	4478
10	M5	4478
11	M3	4413

Q4. What is the average selling price of each BMW model?

SQL Query

```
--Q4. Calculate the average selling price of each BMW model  
SELECT Model, ROUND(AVG(Price_USD), 2) AS Average_Price FROM bmw_sales  
GROUP BY Model ORDER BY Average_Price DESC;
```

Output

	model character varying (50)	average_price numeric
1	7 Series	75570.20
2	3 Series	75566.23
3	i8	75366.27
4	5 Series	75287.84
5	X1	75262.22
6	X3	75016.62
7	M3	74841.59
8	i3	74800.27
9	X5	74708.12
10	M5	74474.93
11	X6	74434.60

Q5. Which regions have the highest number of BMW sales records?

SQL Query

```
--Q5. Find the number of sales records in each region  
SELECT Region, COUNT(*) AS Total_Sales FROM bmw_sales  
GROUP BY Region ORDER BY Total_Sales DESC;
```

Output

	region character varying (50)	total_sales bigint
1	Asia	8454
2	Middle East	8373
3	North America	8335
4	Europe	8334
5	Africa	8253
6	South America	8251

Q6. How many sales records are available for each fuel type?

SQL Query

```
--Q6. Count the number of sales records for each fuel type
SELECT Fuel_Type, COUNT(*) AS Total_Sales FROM bmw_sales
GROUP BY Fuel_Type ORDER BY Total_Sales DESC;
```

Output

	fuel_type character varying (20) 🔒	total_sales bigint 🔒
1	Hybrid	12716
2	Petrol	12550
3	Electric	12471
4	Diesel	12263

Q7. How many sales records are available for each transmission type?

SQL Query

```
--Q7. Count the number of sales records for each transmission type
SELECT Transmission, COUNT(*) AS Total_Sales FROM bmw_sales
GROUP BY Transmission ORDER BY Total_Sales DESC;
```

Output

	transmission character varying (20) 🔒	total_sales bigint 🔒
1	Manual	25154
2	Automatic	24846

Q8. What is the average selling price in each region?

SQL Query

```
--Q8. Calculate the average selling price in each region
SELECT Region, ROUND(AVG(Price_USD), 2) AS Average_Price FROM bmw_sales
GROUP BY Region ORDER BY Average_Price DESC;
```

Output

	region character varying (50)	average_price numeric
1	Asia	75554.93
2	North America	75070.05
3	Europe	74988.36
4	South America	74973.60
5	Africa	74885.77
6	Middle East	74726.79

Q9. What is the total sales volume for each BMW model?

SQL Query

```
--Q9. Calculate the total sales volume for each BMW model
SELECT Model, SUM(Sales_Volume) AS Total_Sales_Volume FROM bmw_sales
GROUP BY Model ORDER BY Total_Sales_Volume DESC;
```

Output

	model character varying (50)	total_sales_volume bigint
1	7 Series	23786466
2	i8	23423891
3	X1	23406060
4	3 Series	23281303
5	i3	23133849
6	5 Series	23097519
7	M5	22779688
8	X3	22745529
9	X5	22709749
10	X6	22661986
11	M3	22349694

Q10. How many sales records are available for each year?

SQL Query

```
--Q10. Count the number of sales records for each year
SELECT Year, COUNT(*) AS Total_Sales FROM bmw_sales
GROUP BY Year ORDER BY Year;
```

Output

	year integer	total_sales bigint
1	2010	3330
2	2011	3278
3	2012	3332
4	2013	3326
5	2014	3350
6	2015	3358
7	2016	3365
8	2017	3290
9	2018	3278
10	2019	3392
11	2020	3205
12	2021	3372
13	2022	3478
14	2023	3219
15	2024	3427

Q11. What is the total revenue generated in each region?

SQL Query

```
--Q11. Calculate the total revenue generated in each region  
SELECT Region, SUM(Price_USD) AS Total_Revenue FROM bmw_sales  
GROUP BY Region ORDER BY Total_Revenue DESC;
```

Output

	region character varying (50)	total_revenue bigint
1	Asia	638741336
2	North America	625708906
3	Middle East	625687400
4	Europe	624952966
5	South America	618607164
6	Africa	618032273

Q12. Which are the top 5 BMW models with the highest average selling price?

SQL Query

```
--Q12. Find the top 5 BMW models with the highest average selling price
SELECT Model, ROUND(AVG(Price_USD), 2) AS Average_Price FROM bmw_sales
GROUP BY Model ORDER BY Average_Price DESC LIMIT 5;
```

Output

	model character varying (50)	average_price numeric
1	7 Series	75570.20
2	3 Series	75566.23
3	i8	75366.27
4	5 Series	75287.84
5	X1	75262.22

Q13. What is the average engine size for each BMW model?

SQL Query

```
--Q13. Calculate the average engine size for each BMW model
SELECT Model, ROUND(AVG(Engine_Size_L), 2) AS Avg_Engine_Size FROM bmw_sales
GROUP BY Model ORDER BY Avg_Engine_Size DESC;
```

Output

	model character varying (50)	avg_engine_size numeric
1	7 Series	3.26
2	i3	3.26
3	M3	3.25
4	M5	3.25
5	3 Series	3.25
6	X6	3.25
7	i8	3.24
8	X5	3.24
9	5 Series	3.24
10	X3	3.23
11	X1	3.23

Q14. What is the average vehicle mileage for each fuel type?

SQL Query

```
--Q14. Calculate the average mileage for each fuel type
SELECT Fuel_Type, ROUND(AVG(Mileage_KM), 2) AS Avg_Mileage FROM bmw_sales
GROUP BY Fuel_Type ORDER BY Avg_Mileage DESC;
```

Output

	fuel_type character varying (20) 🔒	avg_mileage numeric 🔒
1	Diesel	100905.53
2	Electric	100524.27
3	Hybrid	100063.77
4	Petrol	99753.51

Q15. Which regions recorded the highest total sales volume?

SQL Query

--Q15. Calculate the total sales volume for each region

```
SELECT Region, SUM(Sales_Volume) AS Total_Sales_Volume FROM bmw_sales  
GROUP BY Region ORDER BY Total_Sales_Volume DESC;
```

Output

	region character varying (50) 🔒	total_sales_volume bigint 🔒
1	Asia	42974277
2	Europe	42555138
3	North America	42402629
4	Middle East	42326620
5	Africa	41565252
6	South America	41551818

Q16. What is the average sales volume recorded each year?

SQL Query

--Q16. Calculate the average sales volume recorded each year

```
SELECT Year, ROUND(AVG(Sales_Volume), 2) AS Avg_Sales_Volume FROM bmw_sales  
GROUP BY Year ORDER BY Year;
```

Output

	year integer	avg_sales_volume numeric
1	2010	5085.12
2	2011	5112.55
3	2012	5027.58
4	2013	5071.18
5	2014	5062.38
6	2015	5065.58
7	2016	5039.39
8	2017	5051.92
9	2018	5006.79
10	2019	5068.38
11	2020	5089.19
12	2021	5007.31
13	2022	5152.66
14	2023	5053.95
15	2024	5114.63

Q17. What is the average selling price for each fuel type?

SQL Query

```
--Q17. Calculate the average selling price for each fuel type
SELECT Fuel_Type, ROUND(AVG(Price_USD), 2) AS Average_Price FROM bmw_sales
GROUP BY Fuel_Type ORDER BY Average_Price DESC;
```

Output

	fuel_type character varying (20)	average_price numeric
1	Electric	75276.31
2	Diesel	75079.81
3	Petrol	74990.42
4	Hybrid	74797.55

Q18. How many unique BMW models were sold in each region?

SQL Query

```
--Q18. Count the number of unique BMW models sold in each region
SELECT Region, COUNT(DISTINCT Model) AS Total_Models FROM bmw_sales
GROUP BY Region ORDER BY Total_Models DESC;
```

Output

	region character varying (50)	total_models bigint
1	Africa	11
2	Asia	11
3	Europe	11
4	Middle East	11
5	North America	11
6	South America	11

Q19. Which BMW models generate the highest total revenue, ranked from highest to lowest?

SQL Query

```
--Q19. Rank BMW models based on total revenue generated
SELECT Model, SUM(Price_USD * Sales_Volume) AS total_revenue,
       RANK() OVER (ORDER BY SUM(Price_USD * Sales_Volume) DESC) AS revenue_rank
FROM bmw_sales GROUP BY Model;
```

Output

	model character varying (50)	total_revenue bigint	revenue_rank bigint
1	7 Series	1790070249282	1
2	3 Series	1768534028214	2
3	i8	1764743448529	3
4	X1	1752985285361	4
5	5 Series	1735712423092	5
6	i3	1724197530210	6
7	X5	1708653383772	7
8	X3	1707951188482	8
9	M5	1698467355916	9
10	X6	1693423254979	10
11	M3	1667504386622	11

Q20. How does the average selling price of each BMW model compare with the overall average selling price?

SQL Query

```
--Q20. Compare each BMW model's average selling price with the overall average selling price
SELECT
    Model,
    ROUND(AVG(Price_USD), 2) AS model_avg_price,
    ROUND((SELECT AVG(Price_USD) FROM bmw_sales), 2) AS overall_avg_price,
    ROUND(AVG(Price_USD) - (SELECT AVG(Price_USD) FROM bmw_sales), 2) AS price_difference
FROM bmw_sales
GROUP BY Model
ORDER BY model_avg_price DESC;
```

Output

	model character varying (50)	model_avg_price numeric	overall_avg_price numeric	price_difference numeric
1	7 Series	75570.20	75034.60	535.60
2	3 Series	75566.23	75034.60	531.63
3	i8	75366.27	75034.60	331.67
4	5 Series	75287.84	75034.60	253.24
5	X1	75262.22	75034.60	227.62
6	X3	75016.62	75034.60	-17.98
7	M3	74841.59	75034.60	-193.01
8	i3	74800.27	75034.60	-234.33
9	X5	74708.12	75034.60	-326.48
10	M5	74474.93	75034.60	-559.67
11	X6	74434.60	75034.60	-600.00

Key Outcomes

The SQL analysis successfully transformed the prepared BMW sales dataset into meaningful business information by answering key analytical questions through structured queries. Using PostgreSQL, important metrics such as sales volume, average selling price, revenue, regional performance, and model-wise trends were calculated efficiently. The analysis also identified relationships between vehicle characteristics and sales performance, providing valuable insights for business decision-making. Furthermore, the SQL queries served as the foundation for creating DAX measures, KPIs, and interactive visualizations in the Power BI dashboard.

Power BI Dashboards

Overview

The BMW Sales Analytics Dashboard was developed using Microsoft Power BI to transform processed sales data into an interactive and visually appealing business intelligence solution. The dashboard enables users to analyze sales performance, regional trends, vehicle characteristics, and yearly sales patterns through dynamic visualizations and interactive filters.

Dashboard Objectives

- Provide an executive view of BMW sales performance.
- Monitor key performance indicators (KPIs).
- Analyze regional and model-wise sales trends.
- Compare vehicle characteristics such as fuel type, engine size, and transmission.
- Support data-driven business decision-making through interactive visualizations.

Power BI Features Used

Feature	Purpose
KPI Cards	Display key business metrics.
Slicers	Filter data dynamically.
Interactive Visuals	Enable cross-filtering and exploration.
Navigation Buttons	Switch between dashboard pages.
DAX Measures	Calculate dynamic KPIs and metrics.
Multi-Page Dashboard	Organize analysis into separate pages.
Professional Theme	Maintain a clean, BMW-inspired design.

Dashboard Pages

Page 1 - Home Page

Objective

Provides an introduction to the BMW Sales Analytics Dashboard and serves as the main navigation page for accessing analytical dashboards.



Page 2 - Executive Overview

Objective

Presents a high-level summary of BMW sales performance using key performance indicators and major business metrics.

Visuals

- KPI Cards
- Sales Records by Year
- Total Revenue by Region
- Sales by Fuel Type

Features

- Interactive slicers
- KPI cards
- Cross-filtering



Page 3 – Sales Performance

Objective

Presents a high-level summary of BMW sales performance using key performance indicators and major business metrics.

Visuals

- KPI Cards
- Sales Records by Year
- Total Revenue by Region
- Sales by Fuel Type

Features

- Interactive slicers
- KPI cards
- Cross-filtering



Page 4 - Regional Analysis

Objective

Compares BMW sales performance across different geographical regions.

Visuals

- Revenue by Region
- Average Price by Region
- Sales Volume by Region
- Fuel Type Distribution by Region

Features

- Regional comparison
- Interactive filtering
- Cross-filtering



Page 5 - Vehicle Analysis

Objective

Examines vehicle characteristics such as engine size, mileage, transmission, and fuel type.

Visuals

- Average Engine Size by Fuel Type
- Average Selling Price by Transmission
- Average Mileage by Fuel Type
- Sales Volume by Transmission

Features

- Vehicle comparison
- KPI cards
- Dynamic filters



Page 6 - Vehicle Analysis

Objective

Examines vehicle characteristics such as engine size, mileage, transmission, and fuel type.

Visuals

- Average Engine Size by Fuel Type
- Average Selling Price by Transmission
- Average Mileage by Fuel Type
- Sales Volume by Transmission

Features

- Vehicle comparison
- KPI cards
- Dynamic filters



Page 7 - Time Trend Analysis

Objective

Analyzes yearly trends in sales records, revenue, average selling price, and sales volume from 2010 to 2024.

Visuals

- Total Records by Year
- Average Selling Price by Year
- Total Revenue by Year
- Average Sales Volume by Year

Features

- Year-wise trend analysis
- Interactive filtering
- KPI cards



Key Outcomes

The Power BI dashboard successfully converted the analyzed data into an interactive Business Intelligence solution. Multiple dashboard pages were designed to provide insights into executive performance, sales trends, regional analysis, vehicle characteristics, and time-based comparisons. Interactive slicers, navigation buttons, KPI cards, and dynamic visualizations enabled users to explore the data efficiently from different perspectives. Overall, the dashboard demonstrates how business data can be transformed into actionable insights that support data-driven decision-making through an intuitive and user-friendly interface.

Key Business Insights

1. Executive Overview

- The dashboard summarizes BMW sales performance from 2010–2024 using key performance indicators and interactive visualizations.
- Total revenue generated is 19T, with a total sales volume of 253M units.
- The average selling price is 75.03K, covering 11 BMW models.
- Revenue is distributed relatively evenly across regions, with Asia contributing the highest revenue.

2. Sales Performance

- The 7 Series generated the highest revenue and sales volume among all BMW models.
- The average selling price remains fairly consistent across different models, indicating a balanced pricing strategy.
- Average mileage also shows minimal variation, suggesting similar vehicle usage across models.

3. Regional Analysis

- Asia recorded the highest total revenue, followed closely by Europe and North America.
- Sales volume is distributed fairly evenly across all regions.
- Average selling prices remain consistent across different geographical markets.
- Fuel type distribution appears balanced across all regions without significant dominance.

4. Vehicle Analysis

- The average engine size is similar across all fuel types, ranging from approximately 3.2L to 3.3L.
- Diesel vehicles show slightly higher average mileage than other fuel types.
- Automatic and manual transmission vehicles contribute almost equally to total sales volume.
- The average selling price remains nearly identical for both transmission types.

5. Time Trend Analysis

- Revenue remained relatively stable throughout the analysis period, with slight fluctuations in certain years.
- Sales records and average sales volume showed moderate year-to-year variation.
- Average selling prices remained consistent over time, indicating stable pricing.
- Overall sales performance demonstrates a steady trend across the years.

6. Model Insights

- The 7 Series is the leading model in both revenue generation and sales volume.
- Price differences among BMW models are relatively small.
- Sales records are evenly distributed across the available models.
- The dashboard enables quick comparison of model performance using interactive filters.

Business Recommendations

Based on the dashboard analysis, the following recommendations can support better business decision-making:

1. Focus on High-Performing Models

Models such as the 7 Series and 3 Series demonstrate strong sales performance and revenue contribution. Similar high-performing models can be prioritized in future sales strategies.

2. Strengthen Regional Performance

Since Asia generates the highest revenue, maintaining strong market presence in this region while identifying growth opportunities in lower-performing regions can improve overall performance.

3. Monitor Sales Trends

Regular analysis of yearly sales trends can help identify periods of growth or decline, enabling timely business decisions and improved planning.

4. Optimize Product Portfolio

Comparing model-wise revenue, sales volume, and pricing can help evaluate product performance and support future portfolio planning.

5. Leverage Interactive Business Intelligence

The Power BI dashboard can be used as a decision-support tool for monitoring KPIs, analyzing sales performance, and identifying business trends through interactive filtering.

6. Enhance Future Analytics

Integrating additional business data, such as customer demographics, dealer performance, and marketing campaigns, can provide deeper insights and support more comprehensive analysis.

Challenges Faced

During the development of the BMW Sales Analytics Dashboard, several challenges were encountered that helped strengthen my technical and analytical skills.

1. Data Preparation

Preparing the dataset for analysis required careful validation of data quality, including verifying data types, checking for missing values, duplicates, and ensuring the dataset was properly structured before importing it into PostgreSQL and Power BI.

2. SQL Query Development

Writing advanced SQL queries was initially challenging, particularly in selecting the appropriate SQL functions, aggregation methods, and query logic to extract meaningful business insights from the dataset.

3. DAX Measure Development

Developing DAX measures for KPIs and calculated metrics required understanding Power BI's formula language and applying the correct functions to produce accurate and dynamic results.

4. Dashboard Design and Formatting

Creating a professional and visually appealing dashboard involved multiple iterations to finalize the layout, formatting, color palette, icons, fonts, and overall BMW-inspired theme while maintaining consistency across all pages.

5. Visualization Selection

Selecting the most suitable visualizations for different business scenarios required careful consideration to ensure the data was presented clearly and effectively.

6. Navigation and User Experience

Implementing page navigation, interactive buttons, slicers, and maintaining proper alignment of dashboard elements required careful planning to provide a smooth and user-friendly experience.

7. Dashboard Layout Optimization

Balancing KPI cards, charts, images, spacing, and visual hierarchy without overcrowding the dashboard was one of the most time-consuming aspects of the project.

Future Enhancements

The following enhancements can further improve the BMW Sales Analytics Dashboard:

- Integrate real-time sales data for live dashboard updates.
- Develop machine learning models to predict future sales trends.
- Include customer demographics and dealer-level information for deeper analysis.
- Connect the dashboard to cloud databases for scalable data management.
- Publish the dashboard using Power BI Service for online access and collaboration.
- Add advanced analytics features such as forecasting and anomaly detection.

Conclusion

The BMW Sales Analytics Dashboard successfully demonstrates the complete data analytics workflow by transforming raw sales data into meaningful business insights. The project involved data preparation using Python (Pandas), data storage and analysis using PostgreSQL, and interactive dashboard development using Microsoft Power BI.

The dashboard provides a comprehensive view of BMW sales performance through multiple analytical perspectives, including executive overview, sales performance, regional analysis, vehicle analysis, time trend analysis, and model insights. Interactive KPI cards, filters, and visualizations enable users to efficiently explore the data and gain valuable business insights.

Throughout the project, various technical and design challenges were addressed, including data preparation, SQL query development, DAX calculations, dashboard layout optimization, visualization selection, and user interface design. Overcoming these challenges enhanced practical skills in Data Analytics, Business Intelligence, and Data Visualization.

Overall, this project demonstrates the ability to collect, prepare, analyze, and visualize business data using industry-standard tools. It serves as a complete end-to-end data analytics solution and showcases practical skills in Python, SQL, PostgreSQL, and Power BI for supporting data-driven decision-making.